

# ST542 French Immersion Project Data Preprocessing

## Data Preprocessing

```
# Load libraries
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.2.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readxl)
```

```
# Load the data, giving names to each variable
french_post_survey <- read_excel("data/French_Post_Survey.xlsx", sheet=1,
                                col_names = c("Timestamp", "Email", "Changed_Major",
                                                "Changed_Minor", "Masters", "Feelings",
                                                "Feelings_Factors", "Class_Level",
                                                "Living_Arrangement", "LA_a1",
                                                "LA_a2", "LA_a3", "LA_a4", "LA_b1", "LA_b2",
                                                "LA_b3", "Groupness1", "Groupness2",
                                                "Groupness3", "Groupness4", "Groupness5",
                                                "Phone_Language", "Phone_Keyboard",
```

```

"Screen_Time", "Interact1", "Interact2",
"Interact3", "Interact4", "Interact5",
"Interact6", "Interact7", "Interact8",
"Interact9", "Interact10", "Speaking1",
"Speaking2", "Speaking3",
"Speaking4", "Speaking5",
"Speak_Hours1", "Speak_Hours2",
"Speak_Hours3", "Speak_Hours4",
"Speak_Hours5", "Use_French1", "Use_French2",
"Use_French3", "Use_French4",
"French_Hours1", "French_Hours2",
"French_Hours3", "French_Hours4",
"Read_French1", "Read_French2",
"Read_French3", "Read_French4",
"Read_French5", "Write_French1",
"Write_French2", "Write_French3",
"Write_French4", "Write_French5",
"Listen_French1", "Listen_French2",
"Listen_French3", "Listen_French4",
"Listen_French5", "Speak_French1",
"Speak_French2", "Speak_French3",
"Speak_French4", "Speak_French5",
"Continue_French1", "Continue_French2",
"Continue_French3", "Continue_French4",
"Continue_French5", "Code", "Extra1",
"Extra2", "Extra3"),

skip = 1)

french_post_survey

```

```
# A tibble: 61 x 81
```

|   | Timestamp           | Email        | Changed_Major | Changed_Minor | Masters | Feelings |
|---|---------------------|--------------|---------------|---------------|---------|----------|
|   | <dtm>               | <chr>        | <chr>         | <chr>         | <chr>   | <chr>    |
| 1 | 2023-07-17 04:48:47 | acactony@nc~ | anthropology  | forensics     | n/a     | Content~ |
| 2 | 2023-07-13 17:47:33 | tstroy@ncsu~ | Biology       | French        | N/A     | Stresse~ |
| 3 | 2023-07-14 08:25:27 | lydiagvalen~ | biology       | business      | none    | Tired, ~ |
| 4 | 2023-07-17 21:19:42 | ccernoc@ncs~ | Biomedical E~ | French        | N/A     | Tired, ~ |
| 5 | 2023-07-15 05:45:52 | jpowell9@nc~ | Business Adm~ | cultural lit~ | none    | Calm, C~ |
| 6 | 2023-07-15 12:48:21 | kagay@ncsu.~ | Communicatio~ | n.a           | n.a     | Calm, T~ |
| 7 | 2023-07-13 18:01:06 | srfox4@ncsu~ | Communicatio~ | French        | Unsure  | Relaxed  |
| 8 | 2023-07-13 18:40:20 | jlcorrei@nc~ | Computer Sci~ | French        | Comput~ | Stresse~ |
| 9 | 2023-07-17 13:12:27 | kebierfe@nc~ | criminology   | n/a           | n/a     | Calm, T~ |

```

10 2023-07-13 18:24:14 snbanyas@nc~ Design Studi~ French      N/A      Calm, C~
# i 51 more rows
# i 75 more variables: Feelings_Factors <chr>, Class_Level <chr>,
#   Living_Arrangement <chr>, LA_a1 <chr>, LA_a2 <chr>, LA_a3 <chr>,
#   LA_a4 <chr>, LA_b1 <chr>, LA_b2 <chr>, LA_b3 <chr>, Groupness1 <chr>,
#   Groupness2 <chr>, Groupness3 <chr>, Groupness4 <chr>, Groupness5 <chr>,
#   Phone_Language <chr>, Phone_Keyboard <chr>, Screen_Time <chr>,
#   Interact1 <chr>, Interact2 <chr>, Interact3 <chr>, Interact4 <chr>, ...

```

```

# Subsetting columns we will use in our analysis
french_data <- french_post_survey |>
  select("Timestamp", "Class_Level", "Living_Arrangement",
         "Phone_Language", "Phone_Keyboard", "Screen_Time",
         "Interact1", "Interact2", "Interact3", "Interact4",
         "Interact5", "Interact6", "Interact7", "Interact8",
         "Interact9", "Interact10", "Speaking1", "Speaking2", "Speaking3",
         "Speaking4", "Speaking5", "Speak_Hours1", "Speak_Hours2",
         "Speak_Hours3", "Speak_Hours4", "Speak_Hours5",
         "Use_French1", "Use_French2", "Use_French3", "Use_French4", "French_Hours1",
         "French_Hours2", "French_Hours3", "French_Hours4", "Read_French1",
         "Read_French2", "Read_French3", "Read_French4", "Read_French5",
         "Write_French1", "Write_French2", "Write_French3", "Write_French4",
         "Write_French5", "Listen_French1", "Listen_French2", "Listen_French3",
         "Listen_French4", "Listen_French5", "Speak_French1", "Speak_French2",
         "Speak_French3", "Speak_French4", "Speak_French5", "Continue_French1",
         "Continue_French2", "Continue_French3", "Continue_French4", "Continue_French5")

french_data

```

```
# A tibble: 61 x 59
```

|    | Timestamp<br><dtm>  | Class_Level<br><chr>    | Living_Arrangement<br><chr> | Phone_Language<br><chr> |
|----|---------------------|-------------------------|-----------------------------|-------------------------|
| 1  | 2023-07-17 04:48:47 | Summer Study Abroad B1~ | c. I lived alone ~          | English, Fren~          |
| 2  | 2023-07-13 17:47:33 | Summer Study Abroad B1~ | b. I lived in the~          | English, Fren~          |
| 3  | 2023-07-14 08:25:27 | Summer Study Abroad A1~ | b. I lived in the~          | English, Fren~          |
| 4  | 2023-07-17 21:19:42 | Summer Study Abroad B1~ | b. I lived in the~          | English, Fren~          |
| 5  | 2023-07-15 05:45:52 | Summer Study Abroad B2~ | a. I lived in the~          | French                  |
| 6  | 2023-07-15 12:48:21 | Summer Study Abroad A1~ | b. I lived in the~          | English                 |
| 7  | 2023-07-13 18:01:06 | Summer Study Abroad B2~ | b. I lived in the~          | English                 |
| 8  | 2023-07-13 18:40:20 | Summer Study Abroad C1~ | a. I lived in the~          | English                 |
| 9  | 2023-07-17 13:12:27 | Summer Study Abroad A1~ | b. I lived in the~          | English                 |
| 10 | 2023-07-13 18:24:14 | Summer Study Abroad A2~ | a. I lived in the~          | English, Fren~          |

```
# i 51 more rows
# i 55 more variables: Phone_Keyboard <chr>, Screen_Time <chr>,
#   Interact1 <chr>, Interact2 <chr>, Interact3 <chr>, Interact4 <chr>,
#   Interact5 <chr>, Interact6 <chr>, Interact7 <chr>, Interact8 <chr>,
#   Interact9 <chr>, Interact10 <chr>, Speaking1 <dbl>, Speaking2 <dbl>,
#   Speaking3 <dbl>, Speaking4 <dbl>, Speaking5 <dbl>, Speak_Hours1 <chr>,
#   Speak_Hours2 <chr>, Speak_Hours3 <chr>, Speak_Hours4 <chr>, ...
```

## Preliminary Data Processing

The Screen Time variable has been calculated in another excel file and needs to be added to this dataset.

```
screen_time_data <- read_excel("data/Screen_Time_Data.xlsx", sheet=1,
                               col_names = c("Timestamp", "Email_Address",
                                              "Screen_Time", "ScreenTimeWk1",
                                              "ScreenTimeWk2", "ScreenTimeWk3",
                                              "ScreenTimeWk4", "AvgScreenTime"),
                               skip = 1)

french_data$Screen_Time <- screen_time_data$AvgScreenTime

french_data
```

```
# A tibble: 61 x 59
  Timestamp          Class_Level Living_Arrangement Phone_Language
  <dtm>              <chr>         <chr>              <chr>
1 2023-07-17 04:48:47 Summer Study Abroad B1~ c. I lived alone ~ English, Fren~
2 2023-07-13 17:47:33 Summer Study Abroad B1~ b. I lived in the~ English, Fren~
3 2023-07-14 08:25:27 Summer Study Abroad A1~ b. I lived in the~ English, Fren~
4 2023-07-17 21:19:42 Summer Study Abroad B1~ b. I lived in the~ English, Fren~
5 2023-07-15 05:45:52 Summer Study Abroad B2~ a. I lived in the~ French
6 2023-07-15 12:48:21 Summer Study Abroad A1~ b. I lived in the~ English
7 2023-07-13 18:01:06 Summer Study Abroad B2~ b. I lived in the~ English
8 2023-07-13 18:40:20 Summer Study Abroad C1~ a. I lived in the~ English
9 2023-07-17 13:12:27 Summer Study Abroad A1~ b. I lived in the~ English
10 2023-07-13 18:24:14 Summer Study Abroad A2~ a. I lived in the~ English, Fren~
# i 51 more rows
# i 55 more variables: Phone_Keyboard <chr>, Screen_Time <dbl>,
#   Interact1 <chr>, Interact2 <chr>, Interact3 <chr>, Interact4 <chr>,
#   Interact5 <chr>, Interact6 <chr>, Interact7 <chr>, Interact8 <chr>,
```

```
# Interact9 <chr>, Interact10 <chr>, Speaking1 <dbl>, Speaking2 <dbl>,
# Speaking3 <dbl>, Speaking4 <dbl>, Speaking5 <dbl>, Speak_Hours1 <chr>,
# Speak_Hours2 <chr>, Speak_Hours3 <chr>, Speak_Hours4 <chr>, ...
```

In the class level variable there are some FLF 201 - Intermediate French 1 and FLF 202 - Intermediate French 2 entries. These are incorrectly entered data, so these values will have to be removed before we begin processing the rest of the data.

```
# All Class Level entries and their counts
```

```
french_data |>
  group_by(Class_Level) |>
  summarize(count = n())
```

```
# A tibble: 7 x 2
```

|   | Class_Level<br><chr>                        | count<br><int> |
|---|---------------------------------------------|----------------|
| 1 | FLF 201                                     | 1              |
| 2 | FLF 202                                     | 2              |
| 3 | Summer Study Abroad A1-level language class | 13             |
| 4 | Summer Study Abroad A2-level language class | 20             |
| 5 | Summer Study Abroad B1-level language class | 8              |
| 6 | Summer Study Abroad B2-level language class | 14             |
| 7 | Summer Study Abroad C1-level language class | 3              |

```
# Removing Class Levels with values FLF 201 or 202
```

```
french_data <- french_data |>
  filter(Class_Level != "FLF 201",
         Class_Level != "FLF 202")
```

```
french_data
```

```
# A tibble: 58 x 59
```

|   | Timestamp<br><dtm>  | Class_Level<br><chr>    | Living_Arrangement<br><chr> | Phone_Language<br><chr> |
|---|---------------------|-------------------------|-----------------------------|-------------------------|
| 1 | 2023-07-17 04:48:47 | Summer Study Abroad B1~ | c. I lived alone ~          | English, Fren~          |
| 2 | 2023-07-13 17:47:33 | Summer Study Abroad B1~ | b. I lived in the~          | English, Fren~          |
| 3 | 2023-07-14 08:25:27 | Summer Study Abroad A1~ | b. I lived in the~          | English, Fren~          |
| 4 | 2023-07-17 21:19:42 | Summer Study Abroad B1~ | b. I lived in the~          | English, Fren~          |
| 5 | 2023-07-15 05:45:52 | Summer Study Abroad B2~ | a. I lived in the~          | French                  |
| 6 | 2023-07-15 12:48:21 | Summer Study Abroad A1~ | b. I lived in the~          | English                 |
| 7 | 2023-07-13 18:01:06 | Summer Study Abroad B2~ | b. I lived in the~          | English                 |

```

8 2023-07-13 18:40:20 Summer Study Abroad C1~ a. I lived in the~ English
9 2023-07-17 13:12:27 Summer Study Abroad A1~ b. I lived in the~ English
10 2023-07-13 18:24:14 Summer Study Abroad A2~ a. I lived in the~ English, Fren~
# i 48 more rows
# i 55 more variables: Phone_Keyboard <chr>, Screen_Time <dbl>,
#   Interact1 <chr>, Interact2 <chr>, Interact3 <chr>, Interact4 <chr>,
#   Interact5 <chr>, Interact6 <chr>, Interact7 <chr>, Interact8 <chr>,
#   Interact9 <chr>, Interact10 <chr>, Speaking1 <dbl>, Speaking2 <dbl>,
#   Speaking3 <dbl>, Speaking4 <dbl>, Speaking5 <dbl>, Speak_Hours1 <chr>,
#   Speak_Hours2 <chr>, Speak_Hours3 <chr>, Speak_Hours4 <chr>, ...

```

```

# Showing the updated Class Level entries
french_data |>
  group_by(Class_Level) |>
  summarize(count = n())

```

```

# A tibble: 5 x 2
  Class_Level                                count
  <chr>                                <int>
1 Summer Study Abroad A1-level language class      13
2 Summer Study Abroad A2-level language class      20
3 Summer Study Abroad B1-level language class       8
4 Summer Study Abroad B2-level language class      14
5 Summer Study Abroad C1-level language class       3

```

## Year

The Timestamp variable needs to be converted to just the year. Year is being treated as categorical with two binary variables, one for 2024 and the other 2025. If both are 0, then we have 2023.

```

# Convert the Timestamp variable to a Year
french_data <- french_data |>
  mutate(Year = year(Timestamp),
         Year_2024 = if_else(Year == 2024, 1, 0),
         Year_2025 = if_else(Year == 2025, 1, 0)) |>
  select(Year_2024, Year_2025, everything(), -Timestamp, -Year)

french_data

```

```

# A tibble: 58 x 60

```

```

      Year_2024 Year_2025 Class_Level      Living_Arrangement Phone_Language
      <dbl>      <dbl> <chr>              <chr>              <chr>
1         0         0 Summer Study Abroad B1~ c. I lived alone ~ English, Fren~
2         0         0 Summer Study Abroad B1~ b. I lived in the~ English, Fren~
3         0         0 Summer Study Abroad A1~ b. I lived in the~ English, Fren~
4         0         0 Summer Study Abroad B1~ b. I lived in the~ English, Fren~
5         0         0 Summer Study Abroad B2~ a. I lived in the~ French
6         0         0 Summer Study Abroad A1~ b. I lived in the~ English
7         0         0 Summer Study Abroad B2~ b. I lived in the~ English
8         0         0 Summer Study Abroad C1~ a. I lived in the~ English
9         0         0 Summer Study Abroad A1~ b. I lived in the~ English
10        0         0 Summer Study Abroad A2~ a. I lived in the~ English, Fren~
# i 48 more rows
# i 55 more variables: Phone_Keyboard <chr>, Screen_Time <dbl>,
#   Interact1 <chr>, Interact2 <chr>, Interact3 <chr>, Interact4 <chr>,
#   Interact5 <chr>, Interact6 <chr>, Interact7 <chr>, Interact8 <chr>,
#   Interact9 <chr>, Interact10 <chr>, Speaking1 <dbl>, Speaking2 <dbl>,
#   Speaking3 <dbl>, Speaking4 <dbl>, Speaking5 <dbl>, Speak_Hours1 <chr>,
#   Speak_Hours2 <chr>, Speak_Hours3 <chr>, Speak_Hours4 <chr>, ...

```

```

# Looking at the number of observations with the year 2024
french_data |>
  group_by(Year_2024) |>
  summarize(count = n())

```

```

# A tibble: 2 x 2
  Year_2024 count
    <dbl> <int>
1         0    44
2         1    14

```

```

# Looking at the number of observations with the year 2025
french_data |>
  group_by(Year_2025) |>
  summarize(count = n())

```

```

# A tibble: 2 x 2
  Year_2025 count
    <dbl> <int>
1         0    39
2         1    19

```

## Class Level

Language levels go from A1 - beginner, A2 - Elementary, B1 - Lower Intermediate, B2 - Upper Intermediate, to C1 - Advanced. Therefore this is an ordinal categorical variable that should be converted to numerical from 0 to 4 in that order.

```
# Look at counts of each Class Level
french_data |>
  group_by(Class_Level) |>
  summarize(count = n())
```

```
# A tibble: 5 x 2
  Class_Level                                count
  <chr>                                     <int>
1 Summer Study Abroad A1-level language class    13
2 Summer Study Abroad A2-level language class    20
3 Summer Study Abroad B1-level language class     8
4 Summer Study Abroad B2-level language class    14
5 Summer Study Abroad C1-level language class     3
```

```
# Create function to convert the ordinal categorical Class Level to numerical
class_level_dict <- c("Summer Study Abroad A1-level language class" = 0,
                     "Summer Study Abroad A2-level language class" = 1,
                     "Summer Study Abroad B1-level language class" = 2,
                     "Summer Study Abroad B2-level language class" = 3,
                     "Summer Study Abroad C1-level language class" = 4)

class_level_column <- c("Class_Level")

convert_class_level <- function(col) {
  return(class_level_dict[col])
}
```

```
# Convert the Class Level variable
french_data <- french_data |>
  mutate(across(all_of(class_level_column), convert_class_level)) |>
  select(Year_2024, Year_2025, Class_Level, everything())

french_data
```

```
# A tibble: 58 x 60
```

```

Year_2024 Year_2025 Class_Level Living_Arrangement Phone_Language
<dbl> <dbl> <dbl> <chr> <chr>
1 0 0 2 c. I lived alone in a room, a~ English, Fren~
2 0 0 2 b. I lived in the student dor~ English, Fren~
3 0 0 0 b. I lived in the student dor~ English, Fren~
4 0 0 2 b. I lived in the student dor~ English, Fren~
5 0 0 3 a. I lived in the home of a F~ French
6 0 0 0 b. I lived in the student dor~ English
7 0 0 3 b. I lived in the student dor~ English
8 0 0 4 a. I lived in the home of a F~ English
9 0 0 0 b. I lived in the student dor~ English
10 0 0 1 a. I lived in the home of a F~ English, Fren~
# i 48 more rows
# i 55 more variables: Phone_Keyboard <chr>, Screen_Time <dbl>,
# Interact1 <chr>, Interact2 <chr>, Interact3 <chr>, Interact4 <chr>,
# Interact5 <chr>, Interact6 <chr>, Interact7 <chr>, Interact8 <chr>,
# Interact9 <chr>, Interact10 <chr>, Speaking1 <dbl>, Speaking2 <dbl>,
# Speaking3 <dbl>, Speaking4 <dbl>, Speaking5 <dbl>, Speak_Hours1 <chr>,
# Speak_Hours2 <chr>, Speak_Hours3 <chr>, Speak_Hours4 <chr>, ...

```

## Living Arrangement

The Living Arrangement variable will be grouped to represent whether or not the student lived with a host family. There were several follow up questions based on this variable, which were stored in the variables: LA\_a1, LA\_a2, LA\_a3, LA\_a4, LA\_b1, LA\_b2, LA\_b3. However, due to the limited nature of the observations of data, subdividing the students among these groups creates subgroups that are too small to analyze.

```

# Show the number of observations of each type of Living Arrangement
french_data |>
  group_by(Living_Arrangement) |>
  summarize(count = n())

```

```

# A tibble: 4 x 2
  Living_Arrangement count
  <chr>             <int>
1 a. I lived in the home of a French-speaking family. 23
2 b. I lived in the student dormitory. 32
3 c. I lived alone in a room, an apartment or a house. 2
4 e. I lived in a room, an apartment or a house with others who were not ~ 1

```

```
# Living with a host family will be represented as 1
french_data <- french_data |>
  mutate(Host_Family = if_else(startsWith(Living_Arrangement, "a"),
                                1, 0)) |>
  select(Year_2024, Year_2025, Class_Level, Host_Family, everything(),
         -Living_Arrangement)

french_data
```

```
# A tibble: 58 x 60
  Year_2024 Year_2025 Class_Level Host_Family Phone_Language Phone_Keyboard
    <dbl>    <dbl>    <dbl>    <dbl> <chr>              <chr>
1         0         0         2         0 0 English, French, ~ English, Fren~
2         0         0         2         0 0 English, French   English, Fren~
3         0         0         0         0 0 English, French   English, Fren~
4         0         0         2         0 0 English, French   English, Fren~
5         0         0         3         1 1 French            English, Fren~
6         0         0         0         0 0 English            English, Fren~
7         0         0         3         0 0 English            English, Fren~
8         0         0         4         1 1 English            English, Fren~
9         0         0         0         0 0 English            English, Fren~
10        0         0         1         1 1 English, French   English, Fren~
# i 48 more rows
# i 54 more variables: Screen_Time <dbl>, Interact1 <chr>, Interact2 <chr>,
#   Interact3 <chr>, Interact4 <chr>, Interact5 <chr>, Interact6 <chr>,
#   Interact7 <chr>, Interact8 <chr>, Interact9 <chr>, Interact10 <chr>,
#   Speaking1 <dbl>, Speaking2 <dbl>, Speaking3 <dbl>, Speaking4 <dbl>,
#   Speaking5 <dbl>, Speak_Hours1 <chr>, Speak_Hours2 <chr>,
#   Speak_Hours3 <chr>, Speak_Hours4 <chr>, Speak_Hours5 <chr>, ...
```

## Phone Language and Phone Keyboard

Both variables will be converted to represent whether or not the students used French on their phone. This will be a binary variable.

```
# Lets look at the values in Phone Language
french_data |>
  group_by(Phone_Language) |>
  summarize(count = n())
```

```
# A tibble: 6 x 2
```

|   | Phone_Language           | count |
|---|--------------------------|-------|
|   | <chr>                    | <int> |
| 1 | English                  | 30    |
| 2 | English, French          | 22    |
| 3 | English, French, Spanish | 1     |
| 4 | English, French, spanish | 1     |
| 5 | French                   | 3     |
| 6 | German                   | 1     |

```
# Lets look at the values in Phone Keyboard
french_data |>
  group_by(Phone_Keyboard) |>
  summarize(count = n())
```

```
# A tibble: 10 x 2
  Phone_Keyboard      count
  <chr>            <int>
1 English              9
2 English, French    38
3 English, French, German  1
4 English, French, Korean  1
5 English, French, Spanish  3
6 English, French, Spanish, Greek  1
7 English, French, Spanish, Korean  1
8 English, French, Spanish, Portuguese  1
9 English, French, spanish  1
10 French              2
```

```
# Convert both Phone variables to binary with 0 being no French, and 1 French
binary_phone_dict <- c("English" = 0,
  "English, French" = 1,
  "English, French, German" = 1,
  "English, French, Korean" = 1,
  "English, French, Spanish" = 1,
  "English, French, spanish" = 1,
  "English, French, Spanish, Greek" = 1,
  "English, French, Spanish, Korean" = 1,
  "English, French, Spanish, Portuguese" = 1,
  "English, French, spanish" = 1,
  "French" = 1,
  "German" = 0)
```

```
binary_phone_columns <- c("Phone_Language", "Phone_Keyboard")
```

```
convert_binary_phone <- function(col) {
  return(binary_phone_dict[col])
}
```

```
# Convert the Phone variables
french_data <- french_data |>
  mutate(across(all_of(binary_phone_columns), convert_binary_phone))
```

```
french_data
```

```
# A tibble: 58 x 60
```

|    | Year_2024 | Year_2025 | Class_Level | Host_Family | Phone_Language | Phone_Keyboard |
|----|-----------|-----------|-------------|-------------|----------------|----------------|
|    | <dbl>     | <dbl>     | <dbl>       | <dbl>       | <dbl>          | <dbl>          |
| 1  | 0         | 0         | 2           | 0           | 1              | 1              |
| 2  | 0         | 0         | 2           | 0           | 1              | 1              |
| 3  | 0         | 0         | 0           | 0           | 1              | 1              |
| 4  | 0         | 0         | 2           | 0           | 1              | 1              |
| 5  | 0         | 0         | 3           | 1           | 1              | 1              |
| 6  | 0         | 0         | 0           | 0           | 0              | 1              |
| 7  | 0         | 0         | 3           | 0           | 0              | 1              |
| 8  | 0         | 0         | 4           | 1           | 0              | 1              |
| 9  | 0         | 0         | 0           | 0           | 0              | 1              |
| 10 | 0         | 0         | 1           | 1           | 1              | 1              |

```
# i 48 more rows
```

```
# i 54 more variables: Screen_Time <dbl>, Interact1 <chr>, Interact2 <chr>,
# Interact3 <chr>, Interact4 <chr>, Interact5 <chr>, Interact6 <chr>,
# Interact7 <chr>, Interact8 <chr>, Interact9 <chr>, Interact10 <chr>,
# Speaking1 <dbl>, Speaking2 <dbl>, Speaking3 <dbl>, Speaking4 <dbl>,
# Speaking5 <dbl>, Speak_Hours1 <chr>, Speak_Hours2 <chr>,
# Speak_Hours3 <chr>, Speak_Hours4 <chr>, Speak_Hours5 <chr>, ...
```

```
# Relook at the Phone Language variable counts
```

```
french_data |>
  group_by(Phone_Language) |>
  summarize(count = n())
```

```
# A tibble: 2 x 2
```

```
Phone_Language count
```

```

      <dbl> <int>
1          0    31
2          1    27

```

```

# Relook at the Phone Keyboard variable counts
french_data |>
  group_by(Phone_Keyboard) |>
  summarize(count = n())

```

```

# A tibble: 2 x 2
  Phone_Keyboard count
      <dbl> <int>
1          0     9
2          1    49

```

## Interaction With French People Frequency

These variables all represent how frequently students interact with French people in different contexts. They are from “not available” to “frequently”, so they are ordinal categorical data.

```

# Looking at the values in one of the Interaction variables
french_data |>
  group_by(Interact10) |>
  summarize(count = n())

```

```

# A tibble: 5 x 2
  Interact10 count
  <chr>      <int>
1 frequently (3-5 times/week) 12
2 never                      13
3 not available                25
4 occasionally (1-2 times/week) 6
5 rarely (<1 times/week)       2

```

```

# Function to convert ordinal categorical Interact variables to numerical
ordinal_interact_dict <- c("not available" = 0,
                           "never" = 1,
                           "rarely (<1 times/week)" = 2,
                           "occasionally (1-2 times/week)" = 3,
                           "frequently (3-5 times/week)" = 4)

```

```
ordinal_interact_columns <- c("Interact1", "Interact2", "Interact3",
                              "Interact4", "Interact5", "Interact6",
                              "Interact7", "Interact8", "Interact9",
                              "Interact10")

convert_ordinal_interact <- function(col) {
  return(ordinal_interact_dict[col])
}

# Convert the Interact variables
french_data <- french_data |>
  mutate(across(all_of(ordinal_interact_columns), convert_ordinal_interact))

french_data
```

```
# A tibble: 58 x 60
  Year_2024 Year_2025 Class_Level Host_Family Phone_Language Phone_Keyboard
    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1         0         0         2         0         1         1
2         0         0         2         0         1         1
3         0         0         0         0         1         1
4         0         0         2         0         1         1
5         0         0         3         1         1         1
6         0         0         0         0         0         1
7         0         0         3         0         0         1
8         0         0         4         1         0         1
9         0         0         0         0         0         1
10        0         0         1         1         1         1
# i 48 more rows
# i 54 more variables: Screen_Time <dbl>, Interact1 <dbl>, Interact2 <dbl>,
#   Interact3 <dbl>, Interact4 <dbl>, Interact5 <dbl>, Interact6 <dbl>,
#   Interact7 <dbl>, Interact8 <dbl>, Interact9 <dbl>, Interact10 <dbl>,
#   Speaking1 <dbl>, Speaking2 <dbl>, Speaking3 <dbl>, Speaking4 <dbl>,
#   Speaking5 <dbl>, Speak_Hours1 <chr>, Speak_Hours2 <chr>,
#   Speak_Hours3 <chr>, Speak_Hours4 <chr>, Speak_Hours5 <chr>, ...
```

## Hours Speaking or Using French

These variables represent the number of hours per day a student spoke French or used French in different contexts. They range from “0-1” to “5+”, so they are all ordinal categorical variables.

```
# Look at the values in one of the Speak Hours variables
# These will be the same as in the French Hours variables
french_data |>
  group_by(Speak_Hours1) |>
  summarize(count = n())
```

```
# A tibble: 6 x 2
  Speak_Hours1 count
  <chr>         <int>
1 0-1           22
2 1-2           16
3 2-3            8
4 3-4            6
5 4-5            3
6 5+             3
```

```
# Function to convert ordinal categorical Hour variables to numerical
ordinal_hours_dict <- c("0-1" = 0,
                        "1-2" = 1,
                        "2-3" = 2,
                        "3-4" = 3,
                        "4-5" = 4,
                        "5+" = 5)

ordinal_hours_columns <- c("Speak_Hours1", "Speak_Hours2", "Speak_Hours3",
                           "Speak_Hours4", "Speak_Hours5", "French_Hours1",
                           "French_Hours2", "French_Hours3", "French_Hours4")

convert_ordinal_hours <- function(col) {
  return(ordinal_hours_dict[col])
}
```

```
# Convert Speak Hours and French Hours variables
french_data <- french_data |>
  mutate(across(all_of(ordinal_hours_columns), convert_ordinal_hours))

french_data
```

```
# A tibble: 58 x 60
  Year_2024 Year_2025 Class_Level Host_Family Phone_Language Phone_Keyboard
  <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
```

```

1      0      0      2      0      1      1
2      0      0      2      0      1      1
3      0      0      0      0      1      1
4      0      0      2      0      1      1
5      0      0      3      1      1      1
6      0      0      0      0      0      1
7      0      0      3      0      0      1
8      0      0      4      1      0      1
9      0      0      0      0      0      1
10     0      0      1      1      1      1
# i 48 more rows
# i 54 more variables: Screen_Time <dbl>, Interact1 <dbl>, Interact2 <dbl>,
#   Interact3 <dbl>, Interact4 <dbl>, Interact5 <dbl>, Interact6 <dbl>,
#   Interact7 <dbl>, Interact8 <dbl>, Interact9 <dbl>, Interact10 <dbl>,
#   Speaking1 <dbl>, Speaking2 <dbl>, Speaking3 <dbl>, Speaking4 <dbl>,
#   Speaking5 <dbl>, Speak_Hours1 <dbl>, Speak_Hours2 <dbl>,
#   Speak_Hours3 <dbl>, Speak_Hours4 <dbl>, Speak_Hours5 <dbl>, ...

```

## Response Variables

There are variables that represent a students Perceived Skill in Reading, Writing, Listening, and Speaking in French in different contexts. There are also variables that represent a students Intention to Continue learning French through different means. These all range from “Not sure” to “Completely sure”, therefore they are ordinal categorical variables.

```

# Look at the values in one of the Read French variables
french_data |>
  group_by(Read_French2) |>
  summarize(count = n())

```

```

# A tibble: 5 x 2
  Read_French2    count
  <chr>          <int>
1 Completely sure     4
2 Moderately sure    20
3 Not sure           2
4 Somewhat sure     10
5 Very sure         22

```

```

# Function to convert ordinal categorical Response variables to numerical
ordinal_dict <- c("Not sure" = 0,

```

```

    "Somewhat sure" = 1,
    "Moderately sure" = 2,
    "Very sure" = 3,
    "Completely sure" = 4)

ordinal_columns <- c("Read_French1", "Read_French2", "Read_French3",
  "Read_French4", "Read_French5", "Write_French1",
  "Write_French2", "Write_French3", "Write_French4",
  "Write_French5", "Listen_French1", "Listen_French2",
  "Listen_French3", "Listen_French4", "Listen_French5",
  "Speak_French1", "Speak_French2", "Speak_French3",
  "Speak_French4", "Speak_French5", "Continue_French1",
  "Continue_French2", "Continue_French3", "Continue_French4",
  "Continue_French5")

convert_ordinal <- function(col) {
  return(ordinal_dict[col])
}

```

```

# Convert Response variables
french_data <- french_data |>
  mutate(across(all_of(ordinal_columns), convert_ordinal))

french_data

```

```

# A tibble: 58 x 60
  Year_2024 Year_2025 Class_Level Host_Family Phone_Language Phone_Keyboard
    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1         0         0         2         0         1         1
2         0         0         2         0         1         1
3         0         0         0         0         1         1
4         0         0         2         0         1         1
5         0         0         3         1         1         1
6         0         0         0         0         0         1
7         0         0         3         0         0         1
8         0         0         4         1         0         1
9         0         0         0         0         0         1
10        0         0         1         1         1         1
# i 48 more rows
# i 54 more variables: Screen_Time <dbl>, Interact1 <dbl>, Interact2 <dbl>,
#   Interact3 <dbl>, Interact4 <dbl>, Interact5 <dbl>, Interact6 <dbl>,
#   Interact7 <dbl>, Interact8 <dbl>, Interact9 <dbl>, Interact10 <dbl>,

```

```
# Speaking1 <dbl>, Speaking2 <dbl>, Speaking3 <dbl>, Speaking4 <dbl>,
# Speaking5 <dbl>, Speak_Hours1 <dbl>, Speak_Hours2 <dbl>,
# Speak_Hours3 <dbl>, Speak_Hours4 <dbl>, Speak_Hours5 <dbl>, ...
```

## Scores

Several variables are referring to the same idea with the same scale, but across many different contexts. It makes sense to add these together to create a single score that represent the overall amount. These scores can then effectively be treated as continuous numerical variables.

```
# Add across each category of the variables to create a score
french_data <- french_data |>
  mutate(Interaction_Score = rowSums(across(starts_with("Interact"))),
         Speaking_Score = rowSums(across(starts_with("Speaking"))),
         Speak_Hours_Score = rowSums(across(starts_with("Speak_Hours"))),
         Use_French_Score = rowSums(across(starts_with("Use_French"))),
         French_Hours_Score = rowSums(across(starts_with("French_Hours"))),
         Read_French_Score = rowSums(across(starts_with("Read_French"))),
         Write_French_Score = rowSums(across(starts_with("Write_French"))),
         Listen_French_Score = rowSums(across(starts_with("Listen_French"))),
         Speak_French_Score = rowSums(across(starts_with("Speak_French"))),
         Competence_Score = rowSums(across(c(Read_French_Score,
                                             Write_French_Score,
                                             Listen_French_Score,
                                             Speak_French_Score))),
         Intention_Score = rowSums(across(starts_with("Continue_French")))) |>
  select(Year_2024, Year_2025, Class_Level, Host_Family,
         Phone_Language, Phone_Keyboard, Screen_Time, Interaction_Score,
         Speaking_Score, Speak_Hours_Score, Use_French_Score, French_Hours_Score,
         Read_French_Score, Write_French_Score, Listen_French_Score,
         Speak_French_Score, Competence_Score, Intention_Score)

french_data
```

```
# A tibble: 58 x 18
```

|   | Year_2024 | Year_2025 | Class_Level | Host_Family | Phone_Language | Phone_Keyboard |
|---|-----------|-----------|-------------|-------------|----------------|----------------|
|   | <dbl>     | <dbl>     | <dbl>       | <dbl>       | <dbl>          | <dbl>          |
| 1 | 0         | 0         | 2           | 0           | 1              | 1              |
| 2 | 0         | 0         | 2           | 0           | 1              | 1              |
| 3 | 0         | 0         | 0           | 0           | 1              | 1              |
| 4 | 0         | 0         | 2           | 0           | 1              | 1              |

|    |   |   |   |   |   |   |
|----|---|---|---|---|---|---|
| 5  | 0 | 0 | 3 | 1 | 1 | 1 |
| 6  | 0 | 0 | 0 | 0 | 0 | 1 |
| 7  | 0 | 0 | 3 | 0 | 0 | 1 |
| 8  | 0 | 0 | 4 | 1 | 0 | 1 |
| 9  | 0 | 0 | 0 | 0 | 0 | 1 |
| 10 | 0 | 0 | 1 | 1 | 1 | 1 |

```

# i 48 more rows
# i 12 more variables: Screen_Time <dbl>, Interaction_Score <dbl>,
#   Speaking_Score <dbl>, Speak_Hours_Score <dbl>, Use_French_Score <dbl>,
#   French_Hours_Score <dbl>, Read_French_Score <dbl>,
#   Write_French_Score <dbl>, Listen_French_Score <dbl>,
#   Speak_French_Score <dbl>, Competence_Score <dbl>, Intention_Score <dbl>

```

## Final Export

```

# Create csv file of the Processed French Data
write_csv(french_data, "data/Processed_French_Data.csv")

```